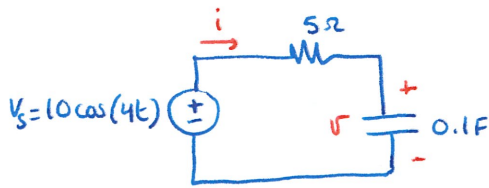


EXERCISE 6

Find $v(t)$ and $i(t)$



First, calculate V_s phasor and impedances

$$V_s = 10 \angle 0^\circ \text{ V}$$

$$Z_R = 5 \Omega$$

$$Z_C = \frac{1}{j\omega C} = -j \frac{1}{\omega C} = -j \frac{1}{4 \times 0.1} = -j2.5 = 2.5 \angle -90^\circ \Omega$$

Simplify the impedances to calculate i :

$$Z_{eq} = Z_R + Z_C = 5 - j2.5$$

Using Ohm's law:

$$I = \frac{V_s}{Z_{eq}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10 \angle 0^\circ}{5.59 \angle -26.56^\circ} = 1.79 \angle 26.56^\circ \text{ A} \implies \underline{i(t) = 1.79 \cos(4t + 26.56^\circ) \text{ A}}$$

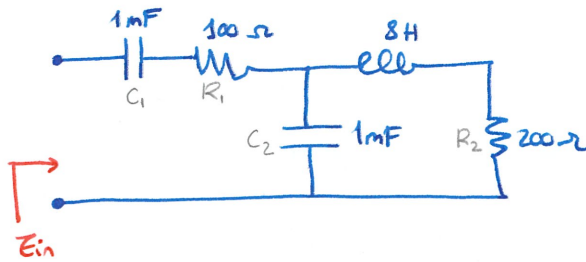
We calculate v using Ohm's law again:

$$V = Z_C \times I = 2.5 \angle -90^\circ \times 1.79 \angle 26.56^\circ = 4.475 \angle -63.44^\circ \text{ V} \implies \underline{v(t) = 4.475 \cos(4t - 63.44^\circ)}$$

Alternatively, use voltage division to find v and then Ohm's law to find i .

EXERCISE 8

Calculate the input impedance if $f = \frac{20}{2\pi} \text{ Hz}$



First, calculate angular frequency and impedances:

$$\omega = 2\pi f = 20 \text{ rad/s}$$

$$Z_{C1} = Z_{C2} = -j \frac{1}{\omega C} = -j \frac{1}{20 \times 0.001} = -j50 \Omega = 50 \angle -90^\circ \Omega$$

$$Z_{R1} = 100 \Omega$$

$$Z_L = j\omega L = j \times 20 \times 8 \text{ H} = j160 \Omega = 160 \angle 90^\circ \Omega$$

$$Z_{R2} = 200 \Omega$$

Then, calculate Z_{in} :

$$Z_{in} = [(Z_{R2} + Z_L) \parallel Z_{C2}] + Z_{R1} + Z_{C1} = [(200 + j160) \parallel (-j50)] + 100 - j50 =$$

$$= \frac{(200 + j160)(-j50)}{200 + j160 - j50} + 100 - j50 = \frac{-j10000 - j^2 160 \times 50}{200 + j110} + 100 - j50 = \frac{8000 - j10000}{200 + j110} + 100 - j50 =$$

$$j^2 = -1$$

$$= \frac{\sqrt{800^2 + 1000^2} \angle \arctan\left(\frac{-1000}{800}\right)}{\sqrt{20^2 + 11^2} \angle \arctan\left(\frac{11}{20}\right)} + 100 - j50 = \frac{1280.62 \angle -51.34^\circ}{22.82 \angle 28.81^\circ} + 100 - j50 = 56.11 \angle -80.15^\circ + 100 - j50 =$$

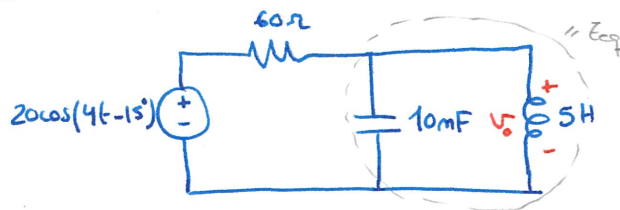
convert to polar
to do the division

$$= 56.11 \cos(-80.15^\circ) + j56.11 \sin(-80.15^\circ) + 100 - j50 = 9.59 - j55.28 + 100 - j50 = \boxed{109.59 - j105.28 \Omega}$$

back to rectangular
to do the addition

EXERCISE 9

Find v_o



First, calculate voltage phasor and impedances:

$$V_s = 20 \angle -15^\circ$$

$$Z_R = 60\Omega$$

$$Z_C = -j \frac{1}{\omega C} = -j \frac{1}{4 \times 0.01} = -j25 = 25 \angle -90^\circ \Omega$$

$$Z_L = j\omega L = j \times 4 \times 5 = j20 = 20 \angle 90^\circ \Omega$$

We calculate Z_{eq} and use voltage division to calculate v_o .

$$Z_{eq} = Z_L \parallel Z_C = j100 = 100 \angle 90^\circ$$

$$V_o = \frac{Z_{eq}}{Z_{eq} + Z_R} \times V_s = \frac{100 \angle 90^\circ}{60 + j100} \times 20 \angle -15^\circ = \frac{2000 \angle 75^\circ}{\sqrt{60^2 + 100^2} \angle \tan^{-1}\left(\frac{100}{60}\right)} = \frac{2000 \angle 75^\circ}{116.62 \angle 59.04^\circ} = 17.15 \angle 15.96^\circ$$

$$\underline{v_o(t) = 17.15 \cos(4t + 15.96^\circ)}$$